In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Main)-2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – V
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	D
2.	C	B	A
3.	C	B	D
4.	C	D	A
5.	D	A	A
6.	A	D	A
7.	B	A	D
8.	D	C	C
9.	D	A	D
10.	A	D	D
11.	D	C	A
12.	A	C	B
13.	B	A	A
14.	A	A	B
15.	B	D	B
16.	B	D	B
17.	C	C	A
18.	B	D	D
19.	A	D	A
20.	C	D	A
21.	B	B	B
22.	A	C	C
23.	D	B	C
24.	A	D	B
25.	D	D	A
26.	A	D	A
27.	D	C	C
28.	B	D	C
29.	D	C	B
30.	D	D	B

Physics

PART – I

SECTION – A

1. **A**

$$e = -\left[\frac{v_y}{v_{0y}}\right]$$

Where v_y and v_{0y} are components of velocities after and before collision along y-axis.

Hence, $e = -\left[\frac{v_y}{-v_0\cos\theta}\right] \quad \Rightarrow \quad v_y = e\, v_0 \cos\theta$

Hence, the required height $= \frac{v_y^2}{2g} = \frac{e^2 v_0^2 \cos^2\theta}{2g}$

2. **C**

Taking the collision between A and C as elastic, it can be concluded that 'C' comes to rest immediately after collision and A starts moving with v_0 (the spring force, if any, during collision can be treated as non-impulsive one.)

At the instant of maximum compression or elongation, relative velocity of A with respect to B would be zero and hence A and B would be moving with a common velocity. From conservation of momentum,

$mv_0 = 2mv \quad \Rightarrow \quad v = v_0/2$

Hence, from conservation of energy

$$(P.E.)_{max} = \frac{1}{2}mv_0^2 - \frac{1}{2}(2m)\left(\frac{v_0}{2}\right)^2$$

$$= \frac{1}{2}mv_0^2 - \frac{1}{4}mv_0^2 = \frac{1}{4}mv_0^2$$

3. **C**

Impact takes place between m_1 and m_2 horizontally. Since m_2 is kept on m_3 and the all the surfaces in contact are smooth, friction does not act between m_2 and m_3 during the displacement of m_3 in the impact. Even though if there is a friction between m_2 and m_3. Which is very less than the impact force, is assumed as non-impulsive force here.

Since the impact between m_1 and m_3 is inelastic, m_1 and m_3 will move combiningly toward right and m_2 does not move due to the absence of friction.

The velocity of the combined mass = $v' = \frac{m_1 v}{m_1 + m_3}$

$$\Rightarrow \quad \frac{|\Delta KE|}{KE} = \frac{\frac{1}{2}m_1v^2 - \frac{1}{2}(m_1+m_3)v'^2}{\frac{1}{2}m_1v^2} = 1 - \left(\frac{m_1+m_3}{m_1}\right)\left(\frac{v'}{v}\right)^2$$

$$= 1 - \left(\frac{m_1+m_3}{m_1}\right)\left(\frac{m_1}{m_1+m_3}\right)^2 = 1 - \frac{m_1}{m_1+m_3} = \frac{m_3}{m_1+m_3}$$

4. **C**

$$KE = (KE_{\text{hollow sphere}}) + (KE_{\text{liquid}}) = (KE_{\text{h.s.}})_{\text{trans}} + (KE)_{\text{rot}} + (KE_{\text{liquid}})_{\text{trans}}$$

$$= \frac{1}{2}(mv^2) + \frac{1}{2}\left(\frac{2}{3}mr^2\right)\left(\frac{v}{r}\right)^2 + \frac{1}{2}mv^2 = \frac{4}{3}mv^2$$

Due to the absence of friction, the liquid does not rotate. It only translates that means it does not possesses rotational K. E.

5. **D**

Velocity of a projectile at any instant of time (t) is

$$v^2 = v_x^2 + v_y^2 = \left(u\cos\theta\right)^2 + \left(u\sin\theta - g\frac{x}{u\cos\theta}\right)^2$$

$$\therefore \qquad KE = \frac{1}{2}mu^2 - mg\times\tan\theta + \frac{gx^2}{u^2\cos^2\theta}$$

The given equation represents equation of a parabola

6. **A**

Conserving angular momentum of rod A about centre of mass.

$$mv\frac{a}{4} = I\omega = \frac{Ma^2}{12}\omega \qquad \Rightarrow \qquad \omega = \frac{3mv}{Ma}$$

7. **B**

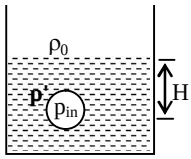
$$\left(\frac{2\pi x_1}{\lambda} - \omega t + \frac{\pi}{6}\right) - \left(\frac{2\pi x_2}{\lambda} - \omega t + \frac{\pi}{8}\right) = 2\pi n$$

$$\Rightarrow x_1 - x_2 = \left(n - \frac{1}{48}\right)\lambda$$

8. **D**

$$P_{in} - p' = \frac{2S}{r}$$

$$\Rightarrow \qquad p_{in} = \frac{2S}{r} + p' = \frac{2S}{r} + p_0 + \rho gH$$



9. **D**

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$, k = spring constant of the uncut spring.

10. **A**

$$f' = \left(\frac{v}{v - v_s}\right)f, \quad f' = \left(\frac{v}{v + v_s}\right)f$$

Beat frequency of f' – f''

11. **D**

$$\Delta W = P_0V_0 + \frac{1}{2}\pi 2P_0V_0 \times \frac{1}{2} = P_0V_0\left(1 + \frac{\pi}{2}\right)$$

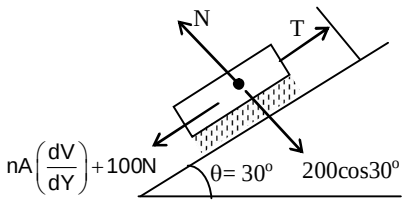
$$\Delta U = \frac{3}{2}\left(P_0 2V_0 - P_0V_0\right) = \frac{3}{2}P_0V_0$$

$$\Delta Q = P_0V_0\left(1 + \frac{\pi}{2} + \frac{3}{2}\right) = P_0V_0\left(\frac{5+\pi}{2}\right)$$

12. **A**

$$\tau = 0.1\times 1m^2 \times \frac{10^{-1}}{10^{-3}} = 10N$$

$$\therefore \quad T = \tau + 100 = 110N$$



13. **B**

Writing the general expression for y ins terms of x as

$y = \frac{1}{1+\left(x-vt^2\right)}$. at t =, $y = 1/\left(1+x^2\right)$. at t = 2s, $y = \frac{1}{1+\left[x-v(2)\right]^2}$

Comparing with the given equation we get 2v = 1 and v= 0.5 m/s.

14. **A**

$$\bar{F}_A = \bar{F}_{AB} + \bar{F}_{AC} = 2\left[\frac{GM^2}{a^2}\right]\cos 30^o = \left[\frac{GM^2}{a^2}\times\sqrt{3}\right]$$

$r = \frac{a}{\sqrt{3}}$. Now, $\frac{Mv^2}{r} = F$

$v = \sqrt{\frac{MG}{a}}$

15. **B**

Apply gas equation

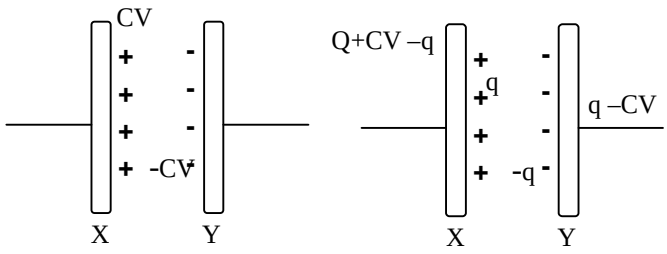
16. **B**

$$\frac{\pi a^2\alpha}{2} = \int_{BOA}\vec{E}.d\vec{l} + \int_{AB}\vec{E}.d\vec{l}$$

$$\frac{\pi a^2\alpha}{2} = \int_{BOA}\vec{E}.d\vec{l} + a^2\alpha \quad \Rightarrow \quad \int_{BOA}\vec{E}.d\vec{l} = a^2\alpha\left(\frac{\pi}{2}-1\right)$$

17. **C**

In the adjacent figure, let X and Y be the positive and negative plates. After charging from the cell, the inner faces of X and Y have charges $\pm$ CV, as shown in (i). The outer surfaces have no charge.



(i) **(ii)**

When charge Q is given to X, let the inner faces of X and Y have charges $\pm$ q. Then, by the principle of charge conservation, the outer faces have charges $(Q+CV-q)$ for X and $(q-CV)$ for Y, as shown in (ii). Now, the outer faces must have equal charges.

$\therefore \quad Q + CV - q = q - CV$

or $\quad 2q = 2CV + Q$

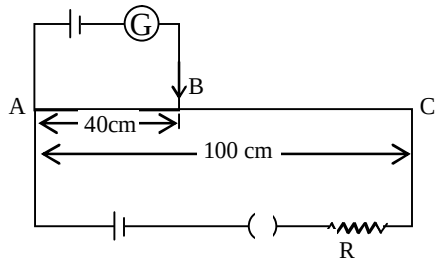
or $\quad q = CV + \frac{Q}{2}$

Potential difference $= \frac{q}{C} = V + \frac{Q}{2C}$

18. **B**

As shown in figure, if R is the unknown resistance, current in the circuit

$$I = \frac{V}{(r+R)} = \frac{2}{(10+R)}$$



Now, as 100 cm wire has resistance 10 Ω, the resistance of 40 cm wire will be

$$40 \times (10/100) = 4\,\text{ohm}.$$

Potential drop across 40 cm wire will be $V = I \times 4$
But V = 10 mV (given)

and $I = \frac{2}{(10+R)} \times 4 \quad \Rightarrow \quad R = 790\,\Omega$

19. **A**

According to given problem,

$$I = \frac{V}{Z} = V / \left[R^2 + \left(1/C^2\omega^2\right)\right]^{\frac{1}{2}} \qquad \text{.....(1)}$$

and $$\frac{I}{2} = \frac{V}{\left[R^2 + \left(9/C^2\omega^2\right)\right]^{\frac{1}{2}}} \qquad \text{.....(2)}$$

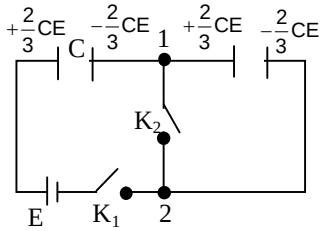
Substituting the value of I from equation (1) in (2),

$$4\left(R^2 + \frac{1}{C^2\omega^2}\right) = R^2 + \frac{9}{C^2\omega^2}, \qquad \text{i.e.,} \qquad \frac{1}{C^2\omega^2} = \frac{3}{5}R^2$$

So than $\frac{X}{R} = \frac{(1/C\omega)}{R} = \frac{\left[(3/5)R^2\right]^{1/2}}{R} = \sqrt{\frac{3}{5}}$

20. **C**

Initially charge configuration (before K_2 is closed) is



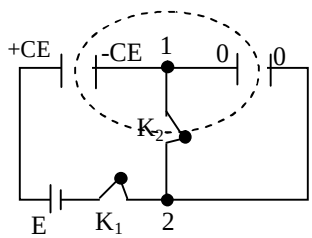
Final charge configuration (after K_2 is closed) is

The net change of charge in the encircled region can happen only because of charge flowing in section 1 → 2.

Initial charge = 0

Final charge = -CE

⇒ Charge flown = -CE from 2 to 1.



21. **B**

Here, one might be tempted to think that since the refractive index of benzene is not given. But actually we do not need this data.

$\mu_1 \sin i_1 = \mu_2 \sin i_2 = \mu_3 \sin i_3$

$\therefore \quad \sin i_3 = \dfrac{\mu_1 \sin i_1}{\mu_3} = \dfrac{3\sqrt{3}}{8}$

22. **A**

Besides the formulae given for prism, we can always use the basic laws of refraction. Using Snell's law for the reaction at AC, we obtain

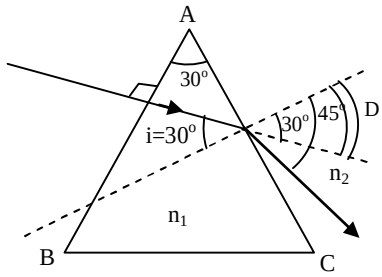
$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$

$$\Rightarrow \quad \frac{\sin 30^o}{\sin r} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad \sin r = 1/\sqrt{2}$$

$$\Rightarrow \quad r = 45^o$$

The angle of deviation, $D = 45^o - 30^o = 15^o$.



23. **D**

Energy of incident photon

$= \dfrac{hc}{\lambda} = \dfrac{12400}{\lambda} eV = \dfrac{12400}{4000} eV = 3.1 eV$

$k_{max} = h\nu - \phi$

$= 3.1 eV - 2.0 eV = 1.1 eV$

Hence, stopping potential 1.1 V

24. **A**

$$\lambda_\alpha = \frac{4}{3R(Z-1)^2}$$

Cut-off wavelengths: $\lambda_1 = \dfrac{hc}{eV_1}; \quad \lambda_2 = \dfrac{hc}{eV_2}$

Using given condition: $(\lambda_2 - \lambda_\alpha) = 3(\lambda_1 - \lambda_\alpha)$

or $\quad 3\lambda_1 - \lambda_2 = 2\lambda_\alpha \qquad$ or $\qquad \dfrac{hc}{e}\left[\dfrac{3}{V_1} - \dfrac{1}{V_2}\right] = \dfrac{8}{3R(Z-1)^2}$

Taking $V_1 = 10 \times 10^3$ V and $V_2 = 20 \times 10^3$ V, we get

$Z = 29$

25. **D**

$$N_1 / N_0 = e^{-\lambda_1 t} \qquad .....(1)$$

$$N_2 / N_0 = e^{-\lambda_1 t} \qquad .....(2)$$

$$\Rightarrow \quad N_1 / N_0 = e^{-(\lambda_1 - \lambda_2)t}$$

Putting $N_1 / N_0 = 1/e$ we obtain $(\lambda_1 - \lambda_2)t = 1$

$$\Rightarrow \quad t = 1/(\lambda_1 - \lambda_2) = 1/(10\lambda - \lambda) = 1/9\lambda$$

26. **A**

$$T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\Rightarrow \quad \frac{\Delta g}{g} = \frac{\Delta l}{T} + 2\frac{\Delta T}{T}$$

$$\Rightarrow \quad \left(\%\frac{\Delta g}{g}\right) = \left(\%\frac{\Delta l}{l}\right) + 2\left(\%\frac{\Delta T}{T}\right)$$

$$\Rightarrow \quad \frac{\Delta g}{g}(\text{in}\%) = 2\% + 2\times 1\% = 4\%$$

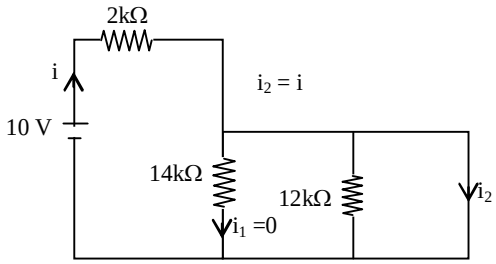
27. **D**

There is no change in λ_{K_α} as it is characteristic X-ray.

28. **B**

$$i = \frac{10}{2} = 5\text{mA} = i_2$$

$$i_1 = 0$$



29. **D**

$$\text{Current gain} = \frac{\Delta I_C}{\Delta I_b} = \beta = \frac{\alpha}{1-\alpha} = 48$$

30. **D**

$$E_{min} = 2\left[\frac{9.1\times 10^{-31}\times\left(3\times 10^8\right)^2}{1.6\times 10^{-13}}\text{MeV} + 0.95\text{ MeV}\right]$$

= 2.924

Chemistry

PART – II

SECTION – A

1. From lines sketches → $n_1 = 2$
$n_2 = 3$
$n_3 = 4$
The radius ratio of given orbit will be
$\Rightarrow r \propto n^2 \Rightarrow 4:9:16$
$\Rightarrow x = 4+9+16$
= 29
$\Rightarrow$ then $\frac{x-4\sigma}{2} \Rightarrow \frac{29-4\times1.5}{2} = \frac{23}{2} = 11.5$

2. $2A \longrightarrow P$
$a_0 - 2x \quad x$
now $\propto = \frac{2x}{a_0} \Rightarrow$ from zero order kintics x = kt
$\propto = \frac{2kt}{a_0}$
Slope = 0.5 $\Rightarrow \frac{2K}{a_0} = .5 \qquad a_0 = .05$
$K = \frac{.5}{2} \times .05 \qquad K = 0.0125$

3. $\underset{\substack{t=0\\t=t}}{H_2A} \underset{\substack{a_0\\a_0-a_0x}}{+ O^{\Theta}H} \rightleftharpoons \underset{\substack{0\\a_0x}}{HA^{\Theta}} + \underset{0}{H_2O}$ (Ka_1)
$\Rightarrow a_0 x = \frac{a_0}{2}$ at 50% neutralization.
From Henderson Equation
$P^H = PK_{a_1} + \log\frac{[HA^-]}{[H_2A]} \quad \Rightarrow [HA^-] = [H_2A]$
$P^H = PK_{a_1}$

5. From 1st law of faraday
$$W = \frac{Mm \times i \times t}{X \times F}$$
$$5_g = \frac{50 \times 965 \times 50}{X \times 96500}$$
X = 5
$$\underset{x=5}{\underbrace{MnO\bar{4} + \overset{\lceil}{M} \longrightarrow Mn^{+2} + \overset{\rceil}{M}^{+5}}}$$
or $MnO_4^- + M + 8H^+ \longrightarrow Mn^{+2} + M^{+5} + 4H_2O$
$\Rightarrow 1+1+8 = 10$

6. $HA \rightleftharpoons H^+ + A^-$

$P^H = 2 \Rightarrow [H^+] = 10^{-2}$

$a_0 \alpha = 10^{-2} \Rightarrow$ Initial conc. = a_0

$\alpha = 0.1$

$\Rightarrow$ Total particle = 1 + α

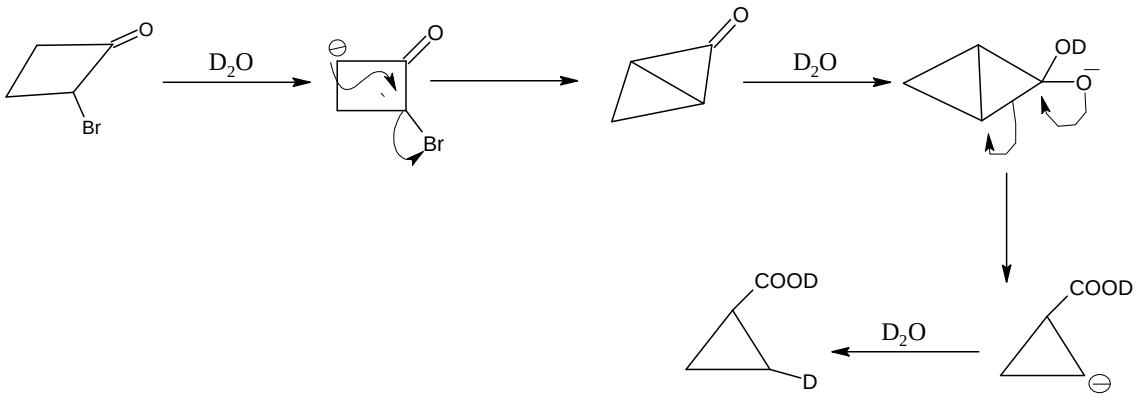
$\pi = (1+\alpha)CST$

$= 0.1 \times 0.0821 \times 298 \times 1.1$

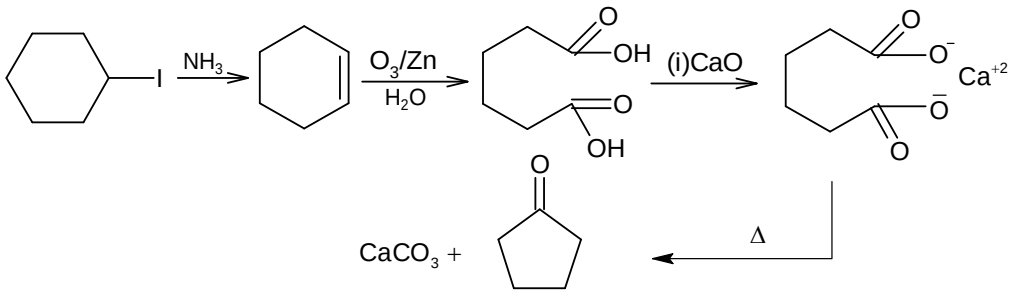
$\pi = 2.69$ atm

7. Lassainge test given by compounds which contains C and N.

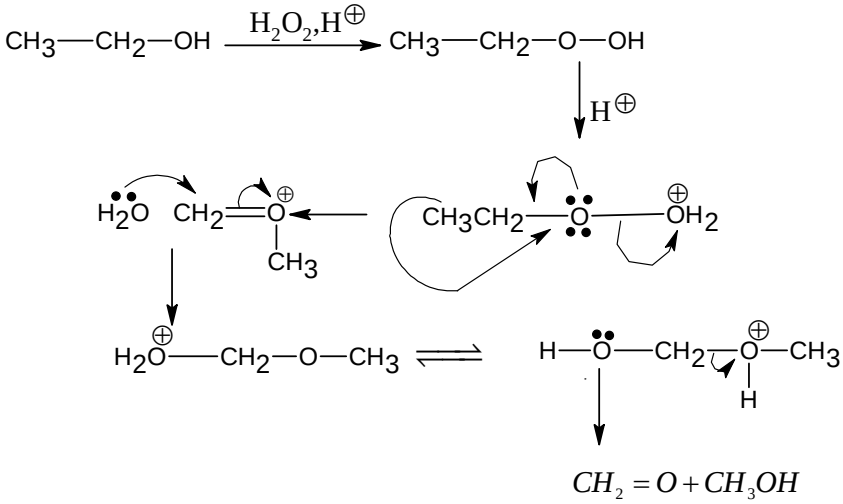
8.



11.



12.



13. Lower the relative humidity better will be drying agent.

$$R.H. = \frac{P_{H_2O} \text{ in equilibrium}}{P_{H_2O} \text{ in atmosphere}}$$

$ACl_2 \cdot 2H_2O$ has lowest relative humidity.

14. In aqueous solution, chromate and dichromate ion exist in equilibrium

$2CrO_4^{-2} + 2H^{\oplus} \rightleftharpoons Cr_2O_7^{-2} + H_2O$

Equilibrium depends depends on both P^H and analytical concentration. CrO_4^{-2} is the predominant in alkaline solutions, but dichromate can become the predominant ion in acidic solution

16. (i) bulky substrate with base gives akene in williamson synthesis

(ii) $R-OH + \overset{\oplus}{H}Br^- \longrightarrow R-\overset{+}{O}H_2 \xrightarrow[-H_2O]{} R^{\oplus}$ (intermidiate)

Br^-

R - Br^-

Greater the stability of $R^{\oplus}$ (Intermediate) Greater will be reactivity of alcohol.

(iii) Basic nature $\propto +I + M$ group

$$\propto \frac{I}{-I, -M \text{ group}}$$

(iv) Dipole moment of Para - chlorodibenzene is zero.

17. $\wedge^{\infty} = \frac{k \times 1000}{5}$ => $\wedge^{\infty} AB = 62 + 76 = 138 \, s\,cm^2\,mol^{-1}$

$$138 = \frac{1.26 \times 10^{-6} \times 1000}{5}$$

s = 9.13×10^{-6} mol/litre.

Number of B^{Θ} per litre = $9.13 \times 10^{-6} \times N_A = 54.96 \times 10^{14}$

18. $S_8 \rightleftharpoons 4S_2$

$1-\alpha \quad 4\alpha$

$\Rightarrow n_T = 1-\alpha+4\alpha = 1+3\alpha$

$\Rightarrow now \frac{r_{mix}(S_2+S_8)}{r_{S_8}} = .75$

$\Rightarrow .75 = \sqrt{\frac{256}{M_{mix}}}$ or $0.75 \times 0.75 = \frac{256}{M_{mix}}$ or $0.5625 = \frac{256}{M_{mix}}$

$\therefore M_{mix} = \frac{256}{0.5625} = 455.11$

$\Rightarrow$ now $\Rightarrow M_{mix} = \frac{256}{1+3\alpha}$

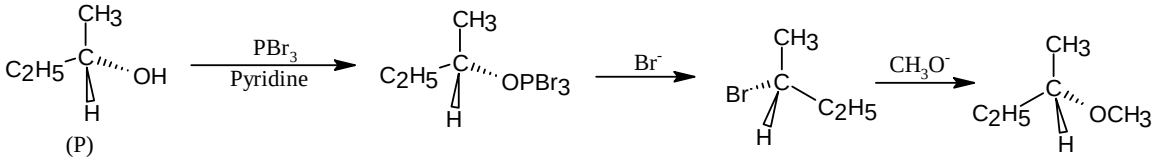
$\Rightarrow 1+3\alpha = .5625$

$3\alpha = 1.5625$

$$\alpha = \frac{1.5625}{3} = 52\%$$

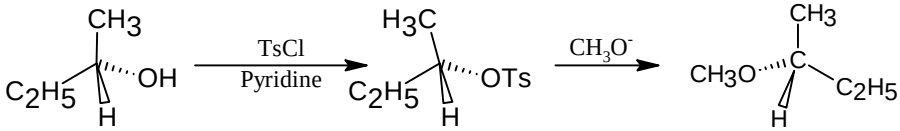
21. $Al(OH)_3$ is not dissolved in excess of the NH_3 solution while Zn $(OH)_2$ dissolved in NH_3.

22.



(P) 'A'

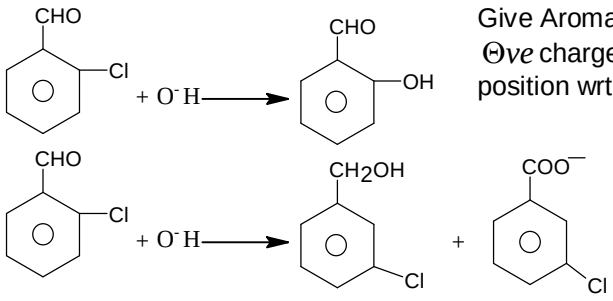
P and A have same configuration.



P B

B have inverted configuration.

23.



Give Aromatic S_N reaction because Θve charge stabilizing group present at ortho position wrt leaving group

28. $$\Delta S = nF\left(\frac{dE^0}{dt}\right)_P$$

$$= 2 \times 96485\, C\, mol^{-1} \times -6 \times 10^{-4} V\, K^{-1}$$

$$= -116 JK^{-1} mol^{-1}$$

29. $$\frac{Tc}{Pc} = \frac{8a}{\frac{27Rb}{a/27b^2}} = \frac{8 \times b}{R}$$

$$= \frac{8 \times .0408}{.0521}$$

$$\simeq 4$$

Mathematics

PART – III

SECTION – A

1. **D**

Put $x-1=t$, then integral become, I = $\int_{-1}^{1}\cos^{-1}\left(t|t|+(t)^3\right)dt$

Applying property then I = $\int_{-1}^{1}\left(\pi-\cos^{-1}\left(t|t|+(t)^3\right)\right)dt$

2. **A**

$2x^2+a^2>x^2+2x+3$

$x^2-2x+a^2-3>0 \qquad \forall \in R.$

hence $4-4\left(a^2-3\right)<0$

$1-a^2+3<0 \Rightarrow a^2>4$

or $a>2$ or $a<-2$

hence $a\in(-\infty,-2)\cup(2,\infty)$.

3. **D**

f(x) is continuous everywhere.

4. **A**

$x>2,$ is a solution

Let $x\le 2$ then $x^2+x+1\ge 4+x^2-4x \Rightarrow x\ge 3/5$

5. **A**

$f(x)=\frac{x+1}{2x-1},$

$f(f(x))=\frac{f(x)+1}{2f(x)-1}=\frac{3x}{3}=x$

6. **A**

Since $3-x^3=|x-3|$ the solution are x = 0, ± 1

Also function is not differentiable at x = 3

7. **D**

$\frac{1}{2}(1+\cos A)+\frac{a}{2}(1+\cos B)=\frac{3c}{2}$

Or $a+b+c=3c \qquad \Rightarrow a+b=2c$

8. **C**

$\tan(A+B)=\frac{-1}{4}\tan A$

9. **D**

$3\sin\theta-4\sin^3\theta=4\sin\theta\sin 2\theta\sin 4\theta$

$\sin\theta = 0 \Rightarrow \theta = n\pi$

Or $3 - 4\sin^2\theta = 4\sin 2\theta \sin 4\theta$

$\Rightarrow \cos 6\theta = -1/2$

10. **D**

Since $I I_1 = a\sec A/2 = 4R\sin A/2$

$I I_2 = a\sec B/2 = 4R\sin B/2$

$I I_3 = a\sec C/2 = 4R\sin C/2$

11. **A**

Since $\dfrac{\tan^2\alpha/2}{\tan^2\beta/2} = 3$

12. **B**

For no trivial solution, we must have

$$\begin{vmatrix} 2a & -2 & 3 \\ 1 & a & 2 \\ 2 & 0 & a \end{vmatrix} = 0 \qquad \Rightarrow a = 2$$

13. **A**

$$\frac{a\cos A + b\cos B + c\cos C}{a+b+c}$$

$$= \frac{R}{2R}\left(\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C}\right) \qquad = 4\sin A/2 \sin B/2 \sin C/2 = \frac{r}{R}$$

14. **B**

$m + n = a\left(\cos\alpha + \sin\alpha\right)^3$

$m - n = a\left(\cos\alpha - \sin\alpha\right)^3$

hence $\left(m+n\right)^{2/3} + \left(m-n\right)^{2/3} = 2a^{2/3}$

15. **B**

$$\left(x+2\right)^{n-1}\left[1 + \frac{x+1}{x+2} + \left(\frac{x+1}{x+2}\right)^2 + \ldots\ldots + \left(\frac{x+1}{x+2}\right)^{n-1}\right] = \left(x+2\right)^{n-1}\left[\frac{1.\left(\left(\frac{x+1}{x+2}\right)^n - 1\right)}{\frac{x+1}{x+2} - 1}\right]$$

$= \left(x+2\right)^n - \left(x+1\right)^n$

Now coefficient of x^r in $\left\{\left(x+2\right)^n - \left(x+1\right)^n\right\}$ is

$= {}^nC_r\, 2^{n-r} - {}^nC_r$

16. **B**

$\left(x\sin\theta+\frac{\cos\theta}{x}\right)^{10}$

$T_{r+1}={}^{10}C_r\left(\sin\theta\right)^{10-r}\left(\cos\theta\right)^r x^{(10-2r)}$

$10-2r=0\Rightarrow r=5$

Hence $f\left(\theta\right)=\frac{{}^{10}C_5}{2^5}\left(\sin 2\theta\right)^5$

17. **A**

$N=\left(5+2\right)^{100}-\left(5-2\right)^{100}$

$=2\{\ {}^{100}C_1 5^{99}\cdot 2^1+{}^{100}C_3 5^{97},2^3+.....+\ {}^{100}C_{99}5.2^{99}]$

hence last three digits are 000

18. **D**

$\tan\left(\frac{A+B}{2}\right)=\frac{\tan A/2+\tan B/2}{1-\tan A/2\tan B/2}$

19. **A**

Let $a_1=a$ and d is common difference $a+a+3d+a+6d=10$

$\Rightarrow 3a+9d=10$

also $a+d+a+9d=8\Rightarrow a+5d=4\Rightarrow 3a+15d=12$

$\therefore d=\frac{1}{3}$ and $a=\frac{7}{3}$

$\therefore S=\frac{20}{2}\left[\frac{14}{3}+\frac{19}{3}\right]=110$

20. **A**

Let $x_1=a-d$ $\qquad y_1=2\left(a-d\right)^2$

$x_2=a$ $\qquad y_2=2a^2$

$x_3=a+d$ $\qquad y_3=2\left(a+d\right)^2$

since y_1,y_2,y_3 are in G.P. then

$4a^4=4\left(a-d\right)^2\left(a+d\right)^2$

or $a^2=\pm\left(a^2-d^2\right)$ $\qquad\Rightarrow 2a^2=d^2$

$a=\pm\frac{d}{\sqrt{2}}$

Now common ratio is $r=\frac{2a^2}{2\left(a-d\right)^2}$ $\qquad r=\frac{1}{\left(3-2\sqrt{2}\right)}=3+2\sqrt{2}$

21. **B**

Let the terms are a,ar,ar^2

then $a\left(1+r+r^2\right)=13$

and $a^2\left(1+r^2+r^4\right)=91$

or $\dfrac{\left(1+r+r^2\right)^2}{\left(1+r^2+r^4\right)}=\dfrac{13}{7} \quad \Rightarrow \dfrac{1+r+r^2}{1+r+r^4}=\dfrac{13}{7}$

hence a = 3, $r-\dfrac{1}{3},3$

$S_n=\dfrac{3\left(3^n-1\right)}{3-1}=\dfrac{3^{n+1}-3}{2}$

22. **C**

Given $\alpha+\beta=\dfrac{1}{\alpha^2}+\dfrac{1}{\beta^2}$ where α,β are the roots, $ab^2+bc^2=2a^2c$

$\Rightarrow {}^{b}\!/_{c},\ {}^{a}\!/_{b},\ {}^{c}\!/_{a}$ are in A.P.

23. **C**

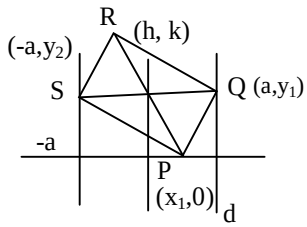
$0 = h + x_1$ …….. (i)

$x_1 = - h$

Let equation of PQ is y = mx + mh

since Q lies on line hence

$y_1 = ma + mh$



$\text{Slope of } QR=-\dfrac{1}{m}=\dfrac{y_1-k}{a-h}\Rightarrow my_1-km=h-a$

or $h+mk=a+\left(a+h\right)m^2$

$\therefore$ Locus is straight line.

24. **B**

$y=mx+\dfrac{1}{m}$

Let P = (h, k) then

$m^2h-mk+1=0 \qquad \Rightarrow m_1+m_2=\dfrac{k}{h},m_1m_2=\dfrac{1}{h}$

Now $\tan\left(\alpha+\beta\right)=\dfrac{\tan\alpha\tan\beta}{1-\tan\left(\alpha+\beta\right)}=1 \qquad \Rightarrow y=x-1$

25. **A**

Let P $\left(3\cos\theta,2\sin\theta\right)$

Equation tangent is $\dfrac{x\cos\theta}{3}+\dfrac{y\sin\theta}{2}=1$

Equation chard A 'P is $y=\dfrac{2\sin\theta}{3\left(\cos\theta+1\right)}\left(x+3\right)$

Now $OM=\dfrac{2\sin\theta}{1+\cos\theta}$

$OQ^2 - MQ^2 = OQ^2 - \left(OQ - OM\right)^2$

Only solving we get, $OQ^2 - MQ^2 = 4$

26. **A**

a and b must have same sigh hence ab > 0

27. **C**

Since $x + \frac{c^4}{x^2} = a^2 \qquad \Rightarrow x^4 - x^2a^2 + c^4 = 0$

Now some of the roots = 0

28. **C**

Since $AA^T = I$

Now $\left(A - I\right)^T = A^T - I$

$A\left(A - I\right)^T = I - A$

$\det\left(A\right)\det\left(A - I\right) = \det\left(I - A\right)$

$\Rightarrow 2\det\left(A - I\right) - \det\left(A - I\right) = 0$

$\Rightarrow \det\left(A - I\right) = 0$

29. **B**

$\frac{dy}{dx} = \frac{\left(x^2 + y^2 + 1\right)}{2xy}$

$\Rightarrow 2xy\ dy = \left(x^2 + 1\right)dx + y^2dx$

$\Rightarrow \frac{2xy\,dy - y^2dx}{x^2} = \left(1 + \frac{1}{x^2}\right)dx$

$\Rightarrow d\left(\frac{y^2}{x}\right) = d\left(x - \frac{1}{x}\right)$

$\Rightarrow \frac{y^2}{x} = x - \frac{1}{x} + c$

$\Rightarrow y^2 + 1 = x^2 + cx$

30. **B**

Total outcomes = 2^n.

Total number of favourable outcomes

$^nC_0 + {}^nC_2 + {}^nC_4 + ... {}^nC_{\left[\frac{n}{2}\right]}$

$= 2^{n-1}$

Thus, required probability = $\frac{2^{n-1}}{2^n} = \frac{1}{2}$.